

Hands-on Assignments: Normalization

Detailed Problem Analysis & Database Decomposition Solutions

Assignment 1: EMPLOYEE Table Analysis

Structure: EMPLOYEE(EMPNO, ENAME, SAL, DEPTNO, DNAME, LOC)

Task: Explain what normal form this table is in and how you can make this into the next normal form.

Current State & Identification:

Assuming all attributes contain single atomic values, the table is in **1st Normal Form (1NF)** and **2nd Normal Form (2NF)** (since the primary key is a single attribute EMPNO, no partial dependencies can exist).

The Problem (Transitive Dependency):

- The Primary Key is **EMPNO**.
- We have the direct functional dependency: **EMPNO → DEPTNO**
- However, department details depend on the department number: **DEPTNO → DNAME, LOC**
- This creates a **Transitive Dependency**: **EMPNO → DEPTNO → (DNAME, LOC)**. Therefore, it violates **3rd Normal Form (3NF)**.

Next Normal Form Solution (Decomposition to 3NF):

To move to 3NF, we split the table to eliminate the transitive relationship:

1. **EMP_DETAILS Table** (Primary Key: EMPNO)

EMP_DETAILS (EMPNO, ENAME, SAL, DEPTNO)

2. **DEPARTMENT Table** (Primary Key: DEPTNO)

DEPARTMENT (DEPTNO, DNAME, LOC)

Assignment 2: STUDENT Table Analysis

Structure: STUDENT(ROLLNO, NAME, AGE, EXAM, MARKS, GRADE)

Task: Explain what normal form this table is in and how you can make this into the next normal form.

Current State & Identification:

This table is currently in **1st Normal Form (1NF)**. The primary key required to identify a unique row must be a composite key: **(ROLLNO, EXAM)** because a student can sit for multiple exams.

The Problem (Partial Dependency):

- The non-prime attributes NAME and AGE depend strictly on ROLLNO alone.
- They do not depend on which EXAM was taken. This creates a **Partial Dependency** (a non-prime attribute depending on a subset of a composite primary key), which violates **2nd Normal Form (2NF)**.
- Additionally, GRADE likely depends directly on MARKS (**MARKS → GRADE**), which represents a transitive dependency to be resolved later.

Next Normal Form Solution (Decomposition to 2NF):

To reach 2NF, we remove the partial dependencies by splitting the student core profile from their exam scores:

1. STUDENT_PROFILE Table (Primary Key: ROLLNO)

STUDENT_PROFILE (ROLLNO, NAME, AGE)

2. EXAM_PERFORMANCE Table (Composite Primary Key: ROLLNO, EXAM)

EXAM_PERFORMANCE (ROLLNO, EXAM, MARKS, GRADE)

Assignment 3: Composite Key EMPLOYEE Table

Structure: EMPLOYEE(EMPNO, PROJECT_NO, NO_OF_DAYS, CUSTOMERNAME)

Note: EMPNO and PROJECT_NO both are together a composite primary key.

Task: Suggest what kind of problem this table has and how to solve the problem.

The Problem (Partial Dependency / 2NF Violation):

- The composite primary key is (EMPNO, PROJECT_NO).
- The attribute NO_OF_DAYS is fully dependent on both the employee and the specific project (EMPNO, PROJECT_NO → NO_OF_DAYS). This is correct.
- However, CUSTOMERNAME belongs strictly to the project itself, meaning it depends entirely on PROJECT_NO (PROJECT_NO → CUSTOMERNAME).
- Because a non-prime attribute depends on only a part of the composite primary key, this is a classic **Partial Dependency**. The table fails 2NF.

How to Solve the Problem (2NF Decomposition):

We solve this anomaly by isolating the project assignment data from the standalone project entity profile:

1. **PROJECT_ASSIGNMENTS Table** (Composite Primary Key: EMPNO, PROJECT_NO)

PROJECT_ASSIGNMENTS (EMPNO, PROJECT_NO, NO_OF_DAYS)

2. **PROJECT_DETAILS Table** (Primary Key: PROJECT_NO)

PROJECT_DETAILS (PROJECT_NO, CUSTOMERNAME)